

Git & GitHub Learning Guide

This guide is designed for beginners who want to learn Git and GitHub for version control, collaboration, and professional software development.

1. Introduction to Git & GitHub

- What is Version Control?
- What is Git?
- What is GitHub?
- Why Developers Use Git

2. Installing and Setup

- Installing Git
- Configuring Username and Email
- Git Bash Basics
- Creating a GitHub Account

3. Git Basics

- `git init`
- `git status`
- `git add`
- `git commit`
- `git log`

4. Working with Branches

- Creating Branches
- Switching Branches
- Merging Branches
- Resolving Merge Conflicts

5. GitHub Basics

- Creating Repositories
- Cloning Repositories

- Pushing Code
- Pulling Changes
- Forking Projects

6. Collaboration Workflow

- Pull Requests
- Code Reviews
- Open Source Contributions
- Team Collaboration

7. Advanced Git Concepts

- git stash
- git reset
- git rebase
- git cherry-pick
- Git Tags

8. Real-World Projects

- Uploading Portfolio Projects
- Managing Java Full Stack Projects
- Version Control for E-Learning Website
- Using GitHub for Resume Building

Recommended 4-Week Learning Roadmap

Week	Focus Area
1	Git Basics + Setup
2	GitHub + Repositories
3	Branching + Collaboration
4	Advanced Git + Real Projects

Common Git Commands:

- git init
- git clone
- git add .
- git commit -m 'message'
- git push origin main

- `git pull origin main`

Git and GitHub are essential tools for every developer. Mastering them will help you collaborate on projects, manage code professionally, and improve your software development workflow.